HEATMAP NODE · PRODUCT REQUIREMENTS AND SPECIFICATIONS

HeatMap Node product requirements and specifications

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DOCUMENT CONTROL

Document ID	HMN-PRS-001	Version	0.8
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/heatmap-node		
Source	docs/03-requirements.md @ b9c67bc		

REVISION HISTORY

Version	Date	Author	Change
0.8	2026-11-01	Amish Chadha	Kit 1.9.1 upgrade: R1 to R15 migrated to PR-001 to PR-015 (renamed from HMN-REQ-001), each traced to a user requirement in HMN-URS-001; Tolerance and Units columns, Was column; PR-016, PR-017 and PR-018 added for the FieldNode core limits and for CAD fidelity and for upkeep; statuses unchanged from HMN-CAL-001 v0.4
0.8	2026-10-02	Amish Chadha	Core above the arm carried into the model (2026-10-02): R7 note, R15 mass 2.08 kg head and 4.92 kg node; no status changes
0.6	2026-10-01	Amish Chadha	Status from HMN-CAL-001 v0.4 after the design for construction (HMN-DDR-003); R13 reported against the value-engineering target
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)

0.3	2026-09-25	Amish Chadha	TRL 3: R13 redefined to the sensor head per HMN-DDR-001 D1; status table replaced by the results of HMN-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

HeatMap Node product requirements and specifications

What the node must do, each requirement traced to a user need in HMN-URS-001 (links in HMN-RTM-001). The requirements describe option A, the fan-aspirated shield with a black globe and a cup anemometer, which is the current design; Amish has not chosen between it and option B (HMN-OPT-001, HMN-DDR-004: proposed, awaiting Amish). They were first written as R1 to R15 and checked by calculation at TRL 3 (HMN-CAL-001 v0.4, after the design for construction, HMN-DDR-003); the column "Was" keeps the old numbers. Targets are not yet validated with users and will be revised after co-design sessions (HMN-PRB-001). Nothing has been tested.

One requirement is not met on paper (PR-009), four are at risk (PR-001, PR-005, PR-010 and PR-014), and one is over its value-engineering target (PR-013). Of the 15 migrated requirements the rest are met on paper or by design; PR-016 and PR-018 are new and to be verified at TRL 4. The decisions behind the targets are HMN-DDR-001 and HMN-DDR-002 (2026-09-25), the budget top-up of 2026-09-26 and the value-engineering decision of 2026-10-01 (HMN-DEC-001): R4 covers 0.8 to 20 m/s with calm intervals flagged, R8 asks for +50 °C at the node, R13 and R15 exclude the FieldNode core and its pole adapter, and `budget_usd` (130) is a hypothetical control target, not a limit.

Table 1. Product requirements and specifications. Status from HMN-CAL-001 v0.4 (`python docs/04-calcs/sizing.py` *), not met first.*

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
PR-009	UR-013	Physical architecture	Measurement height	Sensors at pedestrian height	1.1 to 2.0	range	m	Mounting review with pole owners (TRL 3); side-by-side comparison of 2.0 and 2.8 m (TRL 4)	**Not met**: sensors at 2.67 to 2.69 m pedestrian-height air 0.2 to 0.8 °C warmer in strong sun (HMN-DDR-001 D2 chose 2.8 m for reach and vanda resistance; proposed, awaiting Amish)
PR-005	UR-001, UR-002	Performance	Derived heat stress	Mean radiant temperature and outdoor WBGT computed per ISO 7243 definitions with the Liljegren method; WBGT within the tolerance of a reference WBGT meter	1.0	±, max	°C	Calculation note (TRL 3); collocation with a reference meter (TRL 4)	**At risk**: input error ±0.30 °C RSS of random terms, ±0.55 °C worst case with the fan, plus a one-sided shield bias up to 0.43 ° and height bias up to 0.4 °C; calm intervals flagged Liljegren model error unknown
PR-010	UR-003, UR-004	Usability	Fit to existing poles	Stainless band clamps for 60 to 200 mm round poles; no drilling; fitted from a ladder or lift by two people	30	max	min	Design review (TRL 3); timed trial (TRL 4)	**At risk** (time): 120 degree V-saddle fits 60 to 200 mm; 37 min estimated
PR-014	UR-010	Software	Traceable data	Each node carries a calibration record from CalRig; readings published in an open, documented format with node location and height	Pass or fail	n/a	n/a	Documentation review	**At risk**: the probe calibrated before fitting fits a CalR bay but the assembled 150 mm globe does not; payload format defined
PR-013	UR-007, UR-014	Cost	Sensor head cost	Sensor head (all parts	130	max	USD	Priced BOM, `bom/bom.csv`	**Over target by USD 6.00**: USD

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
				except the FieldNode core and its pole adapter) against the value-engineering target `budget_usd`; a hypothetical control target; cost reported as over or under it. FieldNode core, sun shield and pole adapter are counted against FieldNode; the full-node cost is reported for information					136.00 for the head; core with shield and adapter USD 156.00; full node USD 292.00
PR-001	UR-002	Performance	Air temperature accuracy	Air temperature accuracy in the street against a reference aspirated thermometer, in full sun at wind speeds of 1 m/s or more (1.0 °C below 1 m/s)	0.5	±, max	°C	Shield error estimate (TRL 3); side-by-side field comparison (TRL 4)	**At risk** (restated after the independent review of HMN-STM-001 S6): 0.43 °C with the fan at 1 m/s and more, on an assumed 50 % plate-to-air coupling, a 2 s sensor lag and a s fan run; the energy-balance bound is 0.79 °C above the 0.5 °C target; 1.04 °C a 1 m/s with the fa off
PR-002	UR-002	Performance	Humidity accuracy	Relative humidity accuracy from 20 to 80 %RH at 15 to 40 °C	3	±, max	%RH	Datasheet; CalRig salt fixed points (TRL 4)	Met by design: SHT45 ±1.0 %RH typical

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
PR-003	UR-001, UR-002	Performance	Globe temperature	Standard 150 mm black globe (emissivity about 0.95); probe error after calibration; 90 % response time	0.3 °C; 30 min	±, max; max	°C; min	Calibration in CalRig (TRL 4); response estimate (TRL 3)	Met on paper: probe ±0.18 °C (±0.26 °C with a maximum-tolerance reference); 90 % response about 12 min
PR-004	UR-001	Performance	Wind speed	Wind speed from 0.8 to 20 m/s at ±0.5 m/s or ±10 %, whichever is larger; the share of each interval below the start-up speed is reported so the server can flag biased intervals (restated under HMN-DDR-002; was 0.5 to 20 m/s)	0.8 to 20	range	m/s	Datasheet of the chosen anemometer (TRL 3); wind comparison (TRL 4)	Met on paper: start-up about 0.8 m/s (typical, unverified); calm share flagged at 20 % of an interval
PR-006	UR-006	Software	Reporting	15 min means of all channels by default (adjustable 5 to 60 min); data at the gateway within 30 min; 7 days stored on the node for resend	30	max	min	FieldNode firmware design; airtime calculation (TRL 3)	Met on paper: 20 B payload; 23.7 s/day at SF9; 7 days in 21.5 kB; 16 min latency fo 99 % of intervals
PR-007	UR-005	Electrical	Power autonomy	No mains; days without sun at the default reporting interval	5	min	days	FieldNode energy budget plus the sensor load in HMN-PRC-001 (TRL 3)	Met on paper: sensors and fan 35.4 mW, 35 % o FieldNode's 100 mW design allowance
PR-008	UR-008	Environmental	Outdoor survival	Electronics IP65; operate	-20 to +50	range	°C		Met on paper: FieldNode interio

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
				at the node from -20 to +50 °C (restated under HMN-DDR-002, was +60 °C) with FieldNode's hot-climate sun shield on the core; UV-stable plastics; sensors survive driving rain and dust				Design review; datasheets (TRL 3)	57.2 °C at 50 °C air with its sun shield, against 7 °C; sensor head parts within ratings (shield factor assumed); LiFePO4 charge limit to confirm (STEM S25)
PR-011	UR-008, UR-012	Structural	Wind survival	No failure at a 35 m/s gust; arm deflection at 20 m/s	10	max	mm	Wind load calculation (TRL 3)	Met on paper: arm factor 9.9 or minimum yield (7.5 with the head's weight added); 0.31 mm at 20 m/s; clamp twist factor 2.5 (preload assumed)
PR-012	UR-009	Regulatory	Privacy	Environmental data only: no camera, microphone, Wi-Fi or Bluetooth scanning, or any personal data	Pass or fail	n/a	n/a	Design review of hardware and firmware	Met by design: no imaging, audio or radio scanning parts; environmental payload only
PR-015	UR-012	Physical architecture	Mass and loading on the pole	Mass of the sensor head (restated under HMN-DDR-002, was the complete node); the complete-node mass is reported for the pole owner; no part larger than 300 mm across	4	max	kg	Mass estimate (TRL 3); the D3 model's mass is in `cad/fidelity.json`	Met on paper: sensor head 2.11 kg; complete node 4.98 kg for information; largest part 290 mm

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
				other than the arm					
PR-016	UR-005, UR-008	Electrical	Charging limit of the cell	The FieldNode cell shall not charge above its rated charge temperature, which the interior of the core exceeds at 50 °C air	45	max	°C	Datasheet of the chosen cell; FieldNode firmware (TRL 3 paper)	To verify at TRL 4 interior 57.2 °C a 50 °C air against the usual 45 °C charge limit for LiFePO4 (STEM S25); a cross-repo item for FieldNode, new i this upgrade
PR-017	UR-014	Physical architecture	Buildable from open files	The chosen option shall be modelled to detail level D3 (real stock, every fastener, edge breaks), with a STEP file for each made and bought part and a BOM line for each	Pass or fail	n/a	n/a	`python .kit/cad_check.py --export` (TRL 3)	Met by design: `cad/fidelity.jso` passes at D3 for options A and B
PR-018	UR-011	Usability	Little upkeep	A scheduled visit for option A (clean the shield gaps and the globe, check the fan and the clamp, swap the probe for a calibrated one) shall be needed no more often than the interval below; no consumables other than cable ties and a cloth	6	min	months	Maintenance steps in the build plan (TRL 3 paper); field log of fouling (TRL 4)	To verify at TRL 4 fouling of the gaps and the fan by insects and dust is unknown (STEM S27); option B would need a visit every 5 to 7 days for water (STEM S31b)

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Assumptions

- The FieldNode core performs as stated in FND-CAL-001 (100 mW design sensor allowance, 5.75 days without sun at that load; hot-climate node with the sun shield 2.61 kg, $148.00 after its design for construction, FND-DDR-003).
- WBGT is derived, not measured with a wetted wick, because a natural wet-bulb sensor needs a water reservoir that would need refilling at every node. Option B (HMN-OPT-001) tests that assumption on paper.
- Pole owners permit clamped attachments at the proposed height.
- The mass of the D3 model (4.3 kg for option A in `cad/fidelity.json`) differs from the calculation note's 4.98 kg complete node because the FieldNode core is modelled as an envelope with stand-in internals and the globe wall is drawn at 1.5 mm; the calculation note governs PR-015.